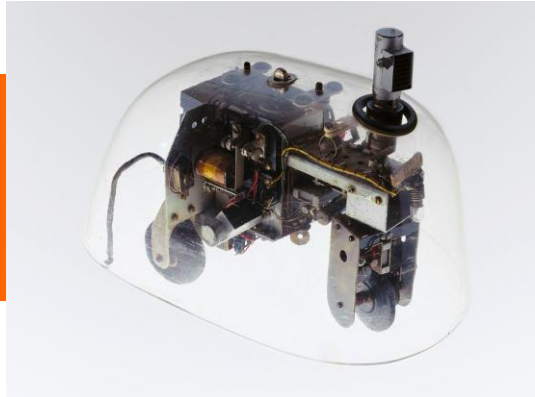
RO2 - Lecture 1
Introduction



we focus on students

Fachhochschule Dortmund
University of Applied Sciences and Arts

1 Introduction

- *Mobile Robots* are:

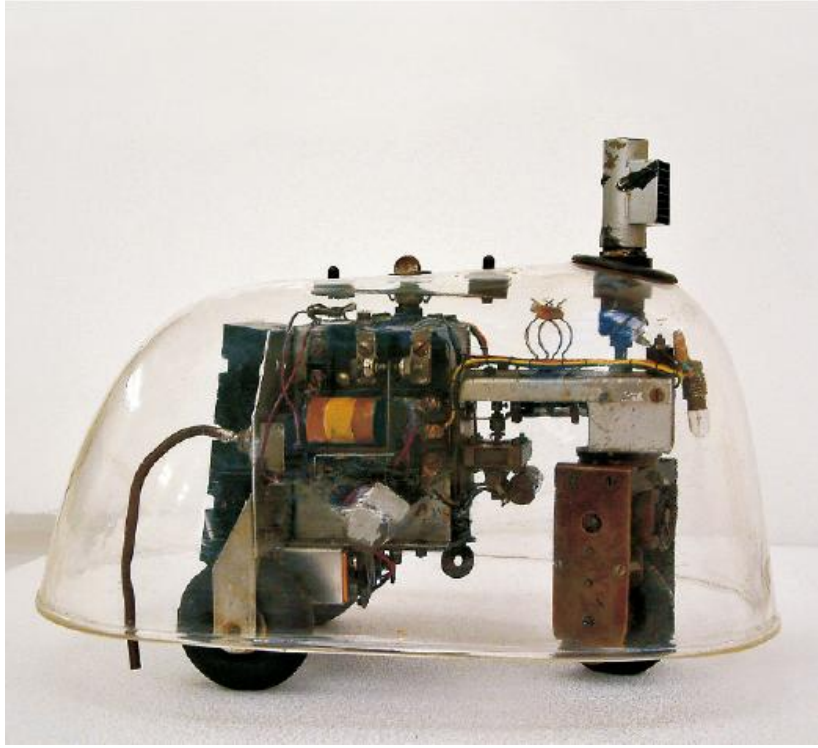
  "A class of robots that are able to move through the environment"

  Peter Corke

- Important aspects of mobile robots:
  - they need to sense their environemnt
  - they need to move from one location to another
- Two main ways how mobile robots move to some place:
  - Simply *react* to what is sensed (e.g. light, ultrasonic measurement)
  - Building a model of the environment, planning movements based on that model

1 Introduction

- Early example of a mobile robot which simply reacts on what is sensed:



Elsie the tortoise, created by British neuroscientist William Grey Walter (1948):

- operated based on immediate stimuli from its environment
- It used simple light and touch sensors to guide its actions
- When it encountered light, it would move towards it, while obstacles would cause it to change direction
- There was no internal representation of the environment or pre-planning of movements.

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1 Introduction

- More recent examples of mobile robots which simply reacts on what is sensed:

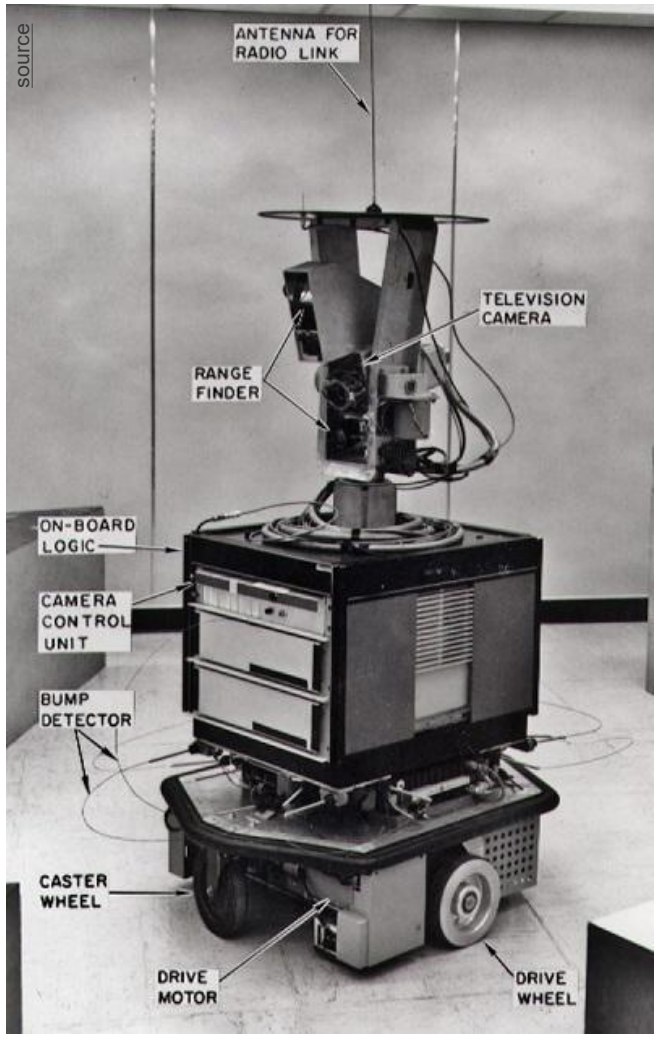
Vacuum Robot Roomba 560 by iRobot (2007)
moves in random patterns, changes direction on hitting an obstacles (detected by bump sensors)

Mower Robot Sileno city by Gardena (2018)

- senses its mowing area through a boundary wire layed around the perimeter of the lawn → this wire emits a low-voltage signal that the mower detects via sensors
- Additionally equipped with collision sensors (bumpers) that help it detect obstacles

1 Introduction

- Early example of a mobile robot which moves according to a model based plan:



Shakey, developed between the mid-1960s and early 1970s at the Stanford Research Institute (SRI)

- used a range of sensors, including:
  - TV camera for vision
  - Bump sensors to detect collisions
  - Sonar range finders to detect distances to objects
- had on-board processors and an antenna for a radio link
- first robot to combine hardware with advanced software that enabled it to plan how to
  - navigate a space
  - move objects and
  - adapt to changes in its environment
- Short video documentation here

1 Introduction

- More recent examples of mobile robots which plan and act:

Vacuum Robot Roomba S9/S9+ by iRobot (2019)
creates and stores maps of the home (based on a camera and a ToF-sensors) and can identify and remember different rooms, allowing users to direct it to clean specific areas using the app or voice commands

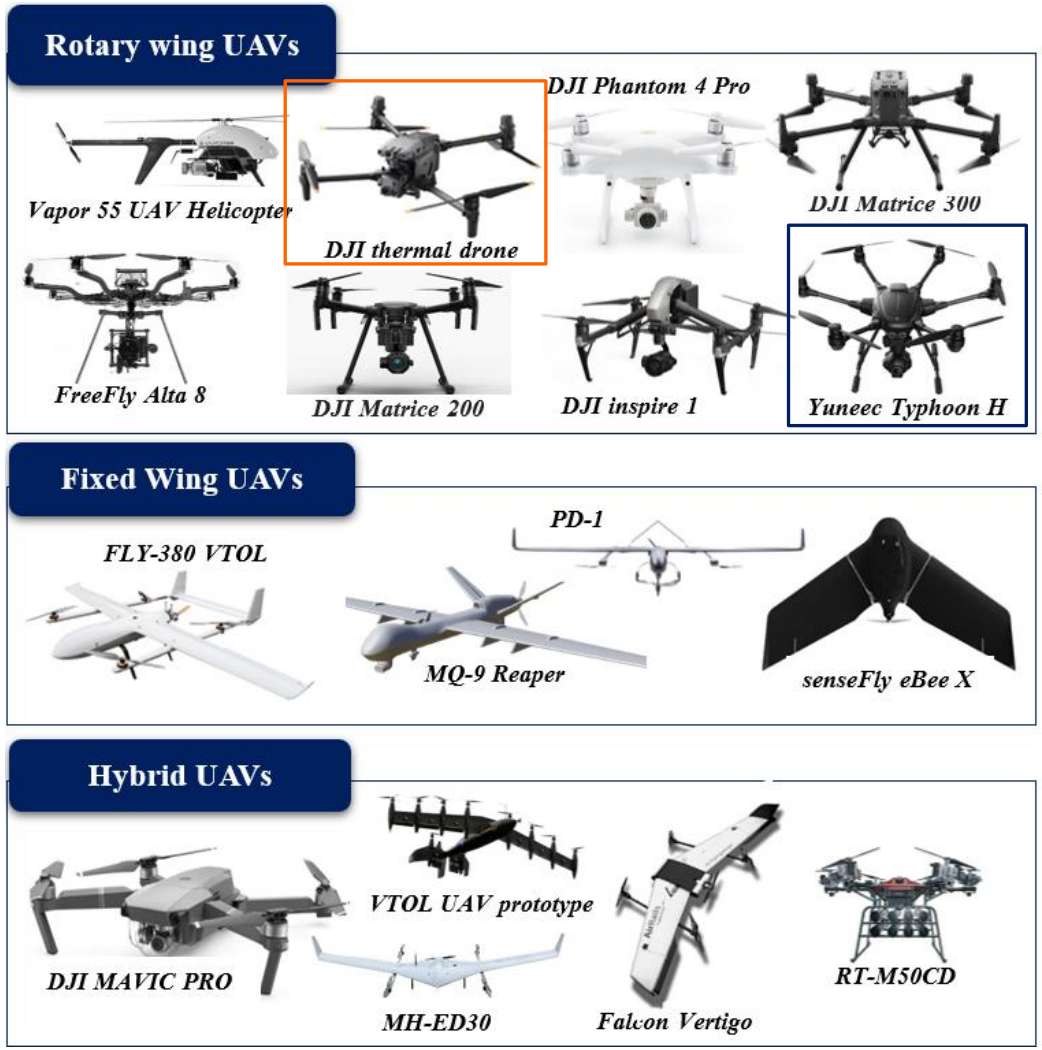
Mower Robot Luba 2 by Mammotion (2023)

- uses virtual boundaries set up via the accompanying app
- uses Real-Time Kinematic (RTK) GPS technology for highly accurate positioning and navigation
- Binocular vision, ultrasonic and bumper for obstacle detection

1 Introduction

- Mobile robots not just limited to ground (and driving via wheels), e.g.:

- Different categories of UAVs based on wing type



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Number of rotatory wings often determines term of subcategory, e.g.:

- 4 rotors → quadcopter
- 6 rotors → hexacopter
- 8 rotors → octocopter

…

1 Introduction

- Higher-level terms used for classification of the whole „UAV system“, including other the platform, the ground control station, data transmission, payload etc: UAS - **U**nmanned **A**erial **S**ystem

Remotely Piloted Aviation Systems - RPAS

Autonomous UAS

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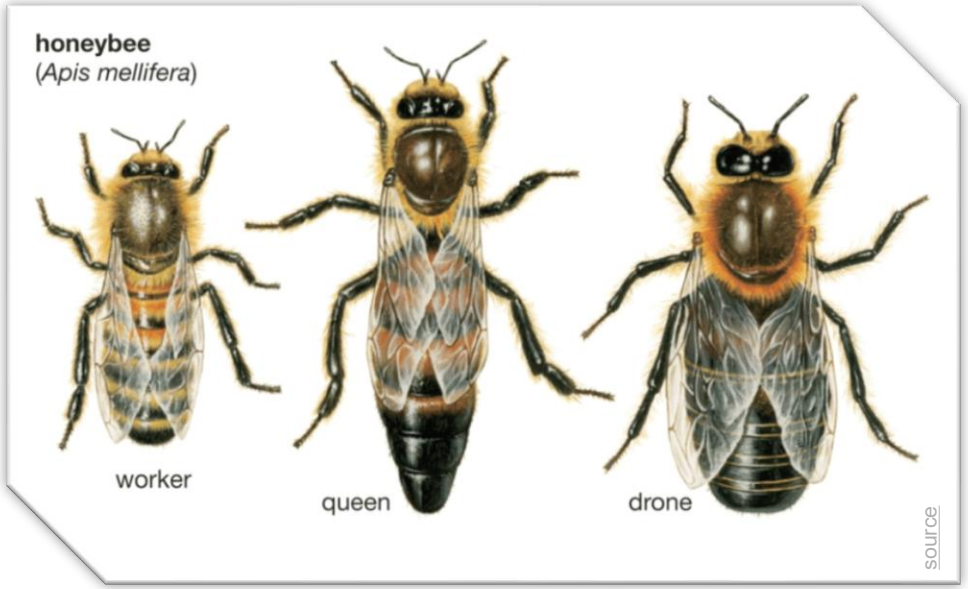
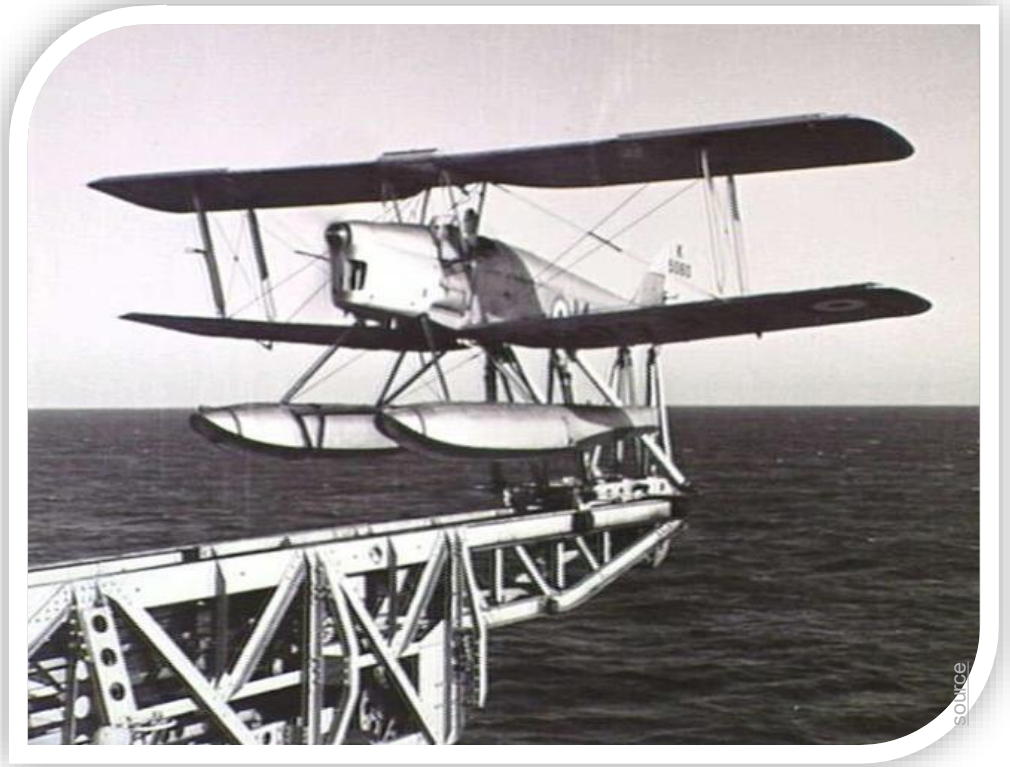
Autonomous UAS

- Origins of the term „Drone“





De Havilland D.H. 82B „Queen Bee“ –
radio controlled target aeroplane (1930s)

1 Introduction

- Use of the term „Drone“ for different UAVs

Consumer/Commercial Drones

Military Combat Drones

Unmanned Combat Aerial Vehicle (UCAV)

Extraterrestrial Drones

Military Reconnaissance Drones

Unmanned Reconnaissance Aerial Vehicle (URAV)

1 Introduction

- Examples for professional application fields of Multirotor Drones

Surveillance and Security

e.g. monitoring large areas, border patrol, crowd control

Aerial Photography and Videography

e.g. film production, journalism, real estate

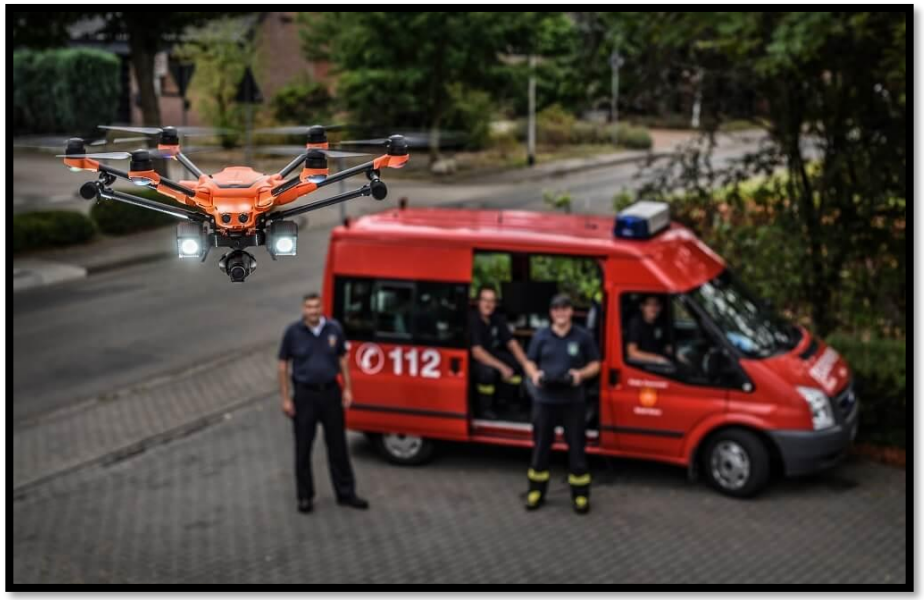
Agriculture

e.g. spreading seeds, pesticides or water; monitoring

- Examples for professional application fields of Multirotor Drones

Delivery Services

Drone delivery systems for delivering small packages to customers



Disaster Management and Search & Rescue

e.g. assist in locating missing persons, assessing damage, and delivering emergency supplies during natural disasters or accidents

1 Introduction

- Scope of this lecture: how do mobile robots, particulary multirotor drones, **sense their environment**?

- How does an autonomously moving mobile robot
  - **avoid collisions**?
  - **know it‘s own pose** (in order to plan moving to a new pose)?

- How are the four rotors of a quadcopter controlled in order to:
  - fly into x, y, or z direction (i.e. **change position**)?
  - change orientiation regarding x, y and z axis (i.e. **change orientation**)?